

Financial Literacy for Students

Porter Clevidence

Antonio Duran-Ramos

Parth Gajjar

Arpit Vora

May 6, 2025

List of Contents

List of Figures	1
List of Tables	2
Project Plan	3
Introduction	3
Team Roles and Responsibilities	3
Methods and Techniques	3
Data	3
Data Description and Analysis	4
Dataset Statistics	4
Text Processing	5
Tokenizer Definition	5
Surface Level Normalization Definition	5
Stopping Definition	5
Stemming/Lemmatizing Definition	5
Index Construction	6
Term Weighting Definition	6
Index Design	6
Retrieval Model	7
Ranking	7
Implementation	8
Source Code	8
Video Demonstration	8
Discussion	9
References	10

List of Figures

List of Tables

Project Plan

Introduction

This project aims to build a search engine that allows students to retrieve relevant documents from a financial dataset. It is part of an educational initiative to promote financial literacy. This enables students to better understand and handle real-world financial situations encountered during their professional lives. To achieve this goal we have designed and implemented an *Information Retrieval (IR)* system that processes textual data, builds an *index* over a document collection, and ranks documents in response to user queries.

Team Roles and Responsibilities

Porter Clevidence

- Implemented BIM search
- Implemented BM25 search
- Implemented Language Model retrieval
- Implemented VSM search
- Search algorithm selection

Antonio Duran-Ramos

- Development environment setup
- Graphical interface design
- LaTeX report setup
-

Parth Gajjar

Arpit Vora

Methods and Techniques

In our system design we chose to use a combination of **Docker** and **uv**, along with various Python libraries. Docker allowed us to standardize our development environment so that our search engine behaved identical, regardless of host machine. **uv** allowed us to

formalize our project as it handles initializing Python virtual environments and it manages dependencies for reproducible Python environments. As for the Python libraries we utilized, **pandas** was used to read from the provided JSON files to create structured data frames. **scikit-learn** was used for english stop words and its functions: **TfidfVectorizer** and **cosine_similarity**. **rank-bm25** was used for its implementation of the BM25 ranking algorithm. **streamlit** was used to implement the graphical interface for our search engine.

Data

The dataset provided consisted of a collection of financial documents, along with static queries and their corresponding relevance judgments.

Data Description and Analysis

Dataset Statistics

The dataset consists of a collection of 5,000 financial text documents, 25 static queries, and corresponding relevance judgments. This dataset was provided in separate JSON files.

Text Processing

Tokenizer Definition

Surface Level Normalization Definition

Stopping Definition

Stemming/Lemmatizing Definition

Index Construction

Term Weighting Definition

Index Design

Retrieval Model

Ranking

Implementation

Source Code

Video Demonstration

Discussion

References

[1]

[2]

[3]